LIGHTWEIGHT NETWORK INTRUSION DETECTION SYSTEM - IDS

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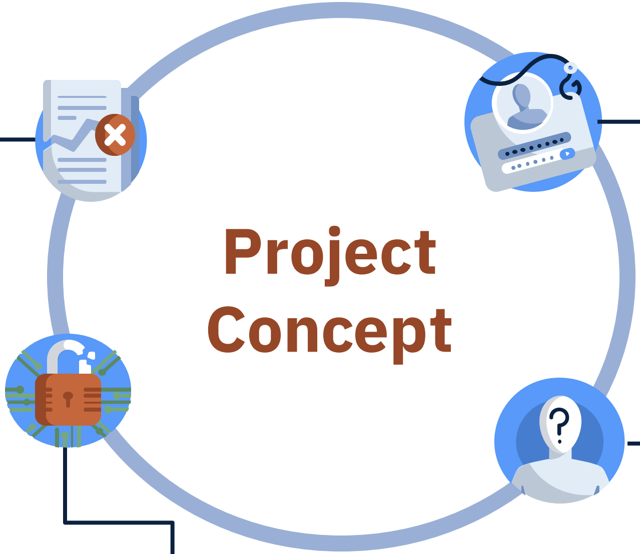
IDS Project Concept

Problem

- Many SMEs (Small and Medium-sized Enterprises) lack resources to monitor network traffic continuously
- Cyber-attacks are increasing in Ireland (O'Donovan, 2025).



Primary Users

- IT administrators
- Network/security analysts
- MSP technicians

Target Customers

- SMEs, schools, clinics, small offices, MSPs (Managed Service Providers)

Solution

- Light IDS using ML to detect suspicious traffic
- Dashboard showing suspicious IPs, ports and alerts

Business Problem & Market Need

Cybersecurity Challenge

- Increasing cyber threats (O'Donovan, 2025).
- SMEs often lack dedicated cybersecurity teams

Operational Impact

- Delayed detection of attacks
- Time spent manually analyzing logs
- Increased operational risk

Business Opportunity

- Affordable security tools for SMEs
- Faster detection using automation

Chosen Technologies

(Python, 2025) (Scikit-learn, 2024)

(Streamlit, 2025)

Python + Scikit-learn

- Used for preprocessing, model training and traffic classification (scikit-learn, 2025)
- Open-source and cost-effective
- Well suited to structured flow-based IDS data
- Compared with MATLAB, it offers lower cost and less vendor dependence

Streamlit

- Used to build the alert dashboard interface
- Open-source and easy to integrate with Python (Streamlit, 2024)
- Helps present ML outputs clearly for non-specialist IT staff
- Compared with Power BI, it is more suitable for a tightly integrated ML prototype

These technologies were selected because they are affordable, flexible, and suitable for SMEs with limited resources.

SWOT Analysis

STRENGTHS

- Lightweight and easy to deploy.
- Automated detection reduces monitoring workload.
- Low implementation cost compared to enterprise solutions

OPPORTUNITIES

- Growing Cybersecurity vendors.
- SMEs lack affordable security tools.
- Increasing regulatory pressure for stronger cybersecurity

WEAKNESSES

- Model retraining required.
- Limited generalization across network environments.
- Dependence on dataset quality and feature selection

THREATS

- Large cybersecurity vendors.
- Integrated enterprise platforms.
- Rapid evolution of cyber-attack techniques

Key Conclusions from the SWOT Analysis

Smarter detection, simpler protection.

- The project has clear commercial potential, particularly for SMEs that have limited cybersecurity resources.
- Its main value is faster detection of suspicious traffic and less manual monitoring.
- The lightweight design and the lower cost of the chosen technologies make it easier for smaller organizations to adopt.
- However, the system has limitations, including the need for retraining, dataset generalization issues and competition from the larger cybersecurity vendors.
- In the end, the project has greater potential as a useful and affordable decision-support security tool, rather than a full enterprise security platform.

References

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